

AVANTHIKA MAKKENA

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Summary

Full-Stack **Software Developer** with **2+ years** of experience in production environments, application design, **cloud solutions**, and **DevOps** practices. Proficient in **Core Java, J2EE, Python, and JavaScript frameworks (Angular, React)**, with a strong background in **RESTful APIs**, web services, **Spring Boot**, and **microservices**. Adept at solution implementation, debugging, and working in Agile/Scrum environments. Skilled in developing scalable systems with Spring Batch, AWS services (EC2, S3, RDS, Lambda, Glue), **Docker**, and **Kubernetes**. Experienced with JPA, XML, SOA principles, and integration using **AWS API Gateway**. Strong experience in **CI/CD** pipelines using **GitHub Actions and Jenkins**. Known for delivering high-quality code, maintaining system reliability, and optimizing performance using APM tools like Grafana and Splunk.

Education

New Jersey Institute of Technology Aug 2023 - Dec 2024

Master of Science, Computer Science

- **Coursework:** Data Structures, Database management systems, Cloud Computing, Machine Learning, Big data analytics

B V Raju Institute of Technology Aug 2018 - Jun 2022

Bachelor of Technology, Computer Science

- **Coursework:** Java Programming, Operating Systems, Data Structures, Software Engineering

Experience

Schneider Electric Jun 2024 – Aug 2024

Software Developer Foxborough, MA

- Developed **RESTful web services** and **APIs** using **Spring Boot** and integrated with **AWS API Gateway**, improving data retrieval speed by 25%.
- Built **production-ready cloud-native pipelines** using **AWS Glue, Lambda, and Athena**.
- Implemented **Spring Batch** for batch data processing and automated workflows.
- Designed automated tests using **Selenium, REST Assured, and Cucumber**, increasing test coverage by 40%; integrated **Sonar** for code quality.
- Deployed containerized **microservices** via **Docker**, improving scalability and deployment speed by 50%.

Schneider Electric Sept 2024 - Dec 2024

Verification & Validation Engineer Harrison, NJ

- Designed **application test strategies** for **energy systems** in **Agile environment**.
- Created **JIRA** stories and documented results in **Confluence**.
- Automated 70% of manual testing using **Selenium, TestNG, and Cucumber**.
- Developed dashboards with **Grafana** for monitoring **system KPIs** in **production environments**

Ensar Solutions Jan 2023 – Jun 2023

Associate Software Engineer – Predictive Case Strategy Platform Hyderabad

- Engineered **ML-powered search APIs** using **Python/Flask**, increasing legal document accuracy by 30%.
- Built and deployed secure **microservices** with **JWT authentication**, reducing unauthorized access by 92%.
- Used **JPA** with optimized **SQL queries** for efficient data handling.
- Designed **Angular UI components** integrated via **RESTful APIs**, improving navigation by 40%.
- Managed **CI/CD** using **GitHub Actions** and containerized deployment with **Docker**.
- Developed services aligned with **SOA**, deployed in a cloud-based **production environment**.

PROJECTS

AI Chat App using LLaMA 2 & Spring Boot

- Developed a local AI-powered chat application using Meta's LLaMA 2 model, Spring Boot, and Java.
- Integrated Ollama for local inference, optimizing performance for low-latency responses.
- Designed robust error handling, and implemented unit tests to ensure 95% code coverage.

URL Shortener – Java, Spring Boot, Redis, MySQL

- Built a scalable URL shortener service, capable of handling 1M+ daily requests.

- Implemented real-time monitoring with Grafana for tracking requests and visualizing performance.
- Deployed on AWS EC2, S3, and RDS for high availability and load balancing.

Healthcare DBMS – Python, Flask, PostgreSQL

- Developed a cloud-based healthcare database managing 50,000+ users with PostgreSQL.
- Designed a user-friendly web interface using Flask, incorporating interactive features for better usability.
- Implemented automated integration and unit tests to ensure data integrity and system reliability.

BookFinder Web App – Flask, MySQL

- Built a full-stack book search app with a responsive UI using Flask and MySQL.
- Implemented efficient query logic and database optimization techniques for faster search results.

Mini Database System (Taco-DB) – C++, SQL

- Developed a custom database system with a multi-layered architecture involving B-tree indexing and query optimization.
- Implemented manual query optimization to reduce query execution time significantly.

Text Chat App – C++, Socket Programming

- Created a TCP-based chat system supporting multiple clients with unicast and broadcast messaging.
- Ensured low-latency, error-free communication under high-load scenarios

Containerized Web Application (Docker, AWS, React)

- Deployed scalable React/Spring Boot microservices via Docker on AWS ECS, ensuring 99.9% uptime.
- Automated CI/CD with GitHub Actions, reducing deployment time by 50%

TECHNICAL SKILLS

- **Languages & Frameworks:** Core Java, Python, JavaScript, TypeScript, Spring Boot, Spring Batch, JPA, Flask, Angular, React
- **Cloud & DevOps:** AWS (Lambda, EC2, S3, RDS, Glue, API Gateway), GCP, Azure, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions
- **Web & APIs:** RESTful APIs, Web Services, Node.js, HTML5, CSS3, XML, Bootstrap
- **Databases:** MySQL, PostgreSQL, MongoDB, Redis, Amazon Neptune Testing: Selenium, REST Assured, JUnit, TestNG, Cucumber, SonarQube, Postman
- **Tools:** Eclipse, Git, Jira, Confluence, Grafana, Splunk, Figma, Bitbucket
- **Methodologies:** Agile/Scrum, CI/CD, SOA, Risk Management, Debugging, Solution Implementation, Application Design

Certifications

- AWS Certified Cloud Practitioner
- Python for Everybody – University of Michigan